Succorfish UW Modem

Succorfish Delphis V3 underwater acoustic modem based on Newcastle University's Nanomodem.

Link to the product: https://succorfish.com/products/delphis/

Updated drivers:

nm3firmwareupdatefiles-20230531.zip

Technical specifications:

delphis-data-sheet_v9_3.2.pdf

User guide:

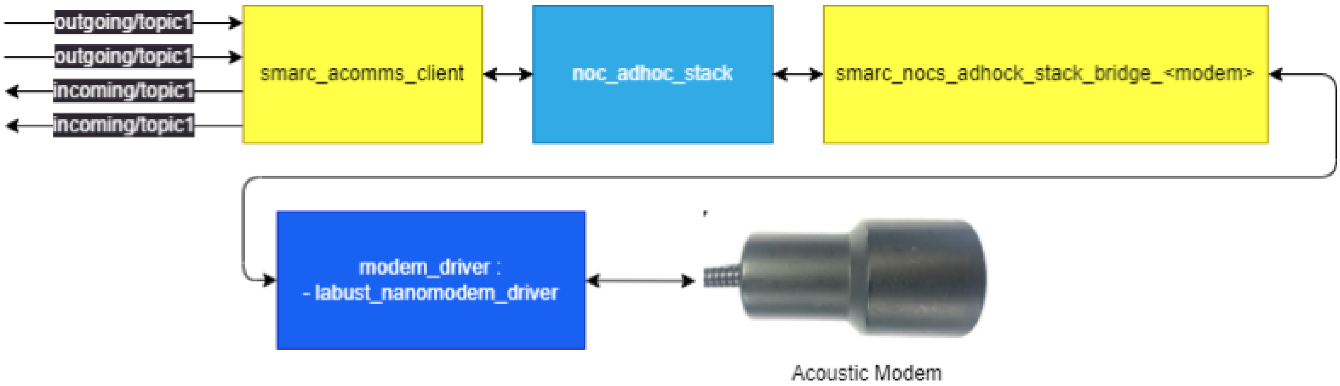
succorfish_delphis_user_guide_v4.pdf

Delphi cable pinout:

delphis_cable_pin_out.pdf

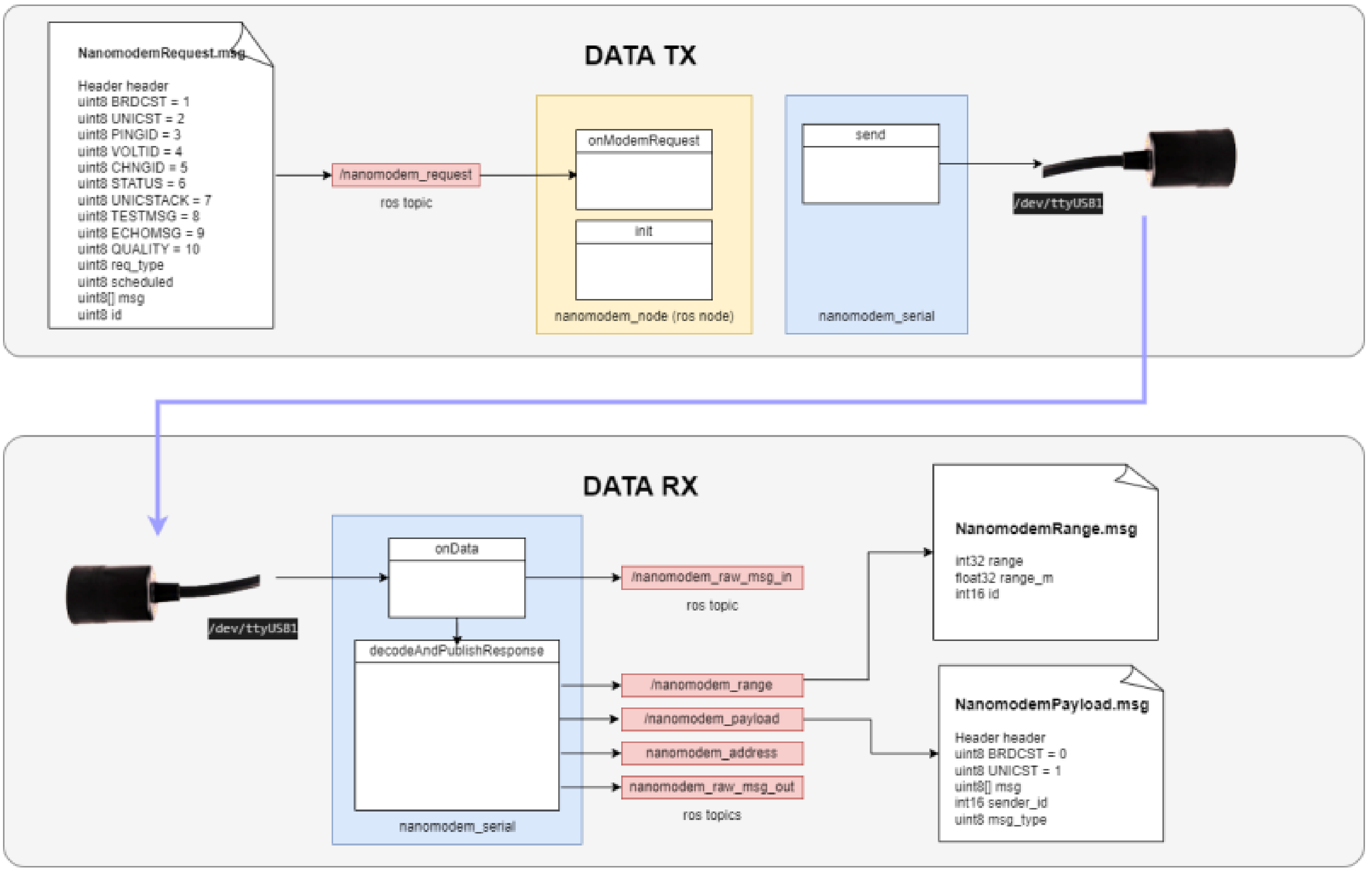
smarc_acomms

Underwater communications for SAM and LoLo, compatible with NanoModem3 acoustic modem : https://github.com/matthew-william-lock/smarc_acomms



LABUST Nanomodem Driver

LABUST has written a `low-level` ROS driver for the Delphis V3. In other words, the driver allows us to use ROS to trigger the same functions as defined in the **user guide**. The driver does not allow for the transmission of arbitrary ROS messages. Shown below is an overview of the driver structure for RX and TX.



Launching ROS Node

```
roslaunch labust_nanomodem nanomodem.xml driver_port:=/dev/ttyUSB0
```

Q&A

Below are some questions we have asked succorfish.

Potting of OEM Module

Q- What material should be used for potting?
A- Polyurethane Potting Compound. Product suggestion:
https://www.acoustics.co.uk/product/aptflex-f21/

Q- How thick should the potting be outside the module?
A- 1-3mm would be the recommended thickness.

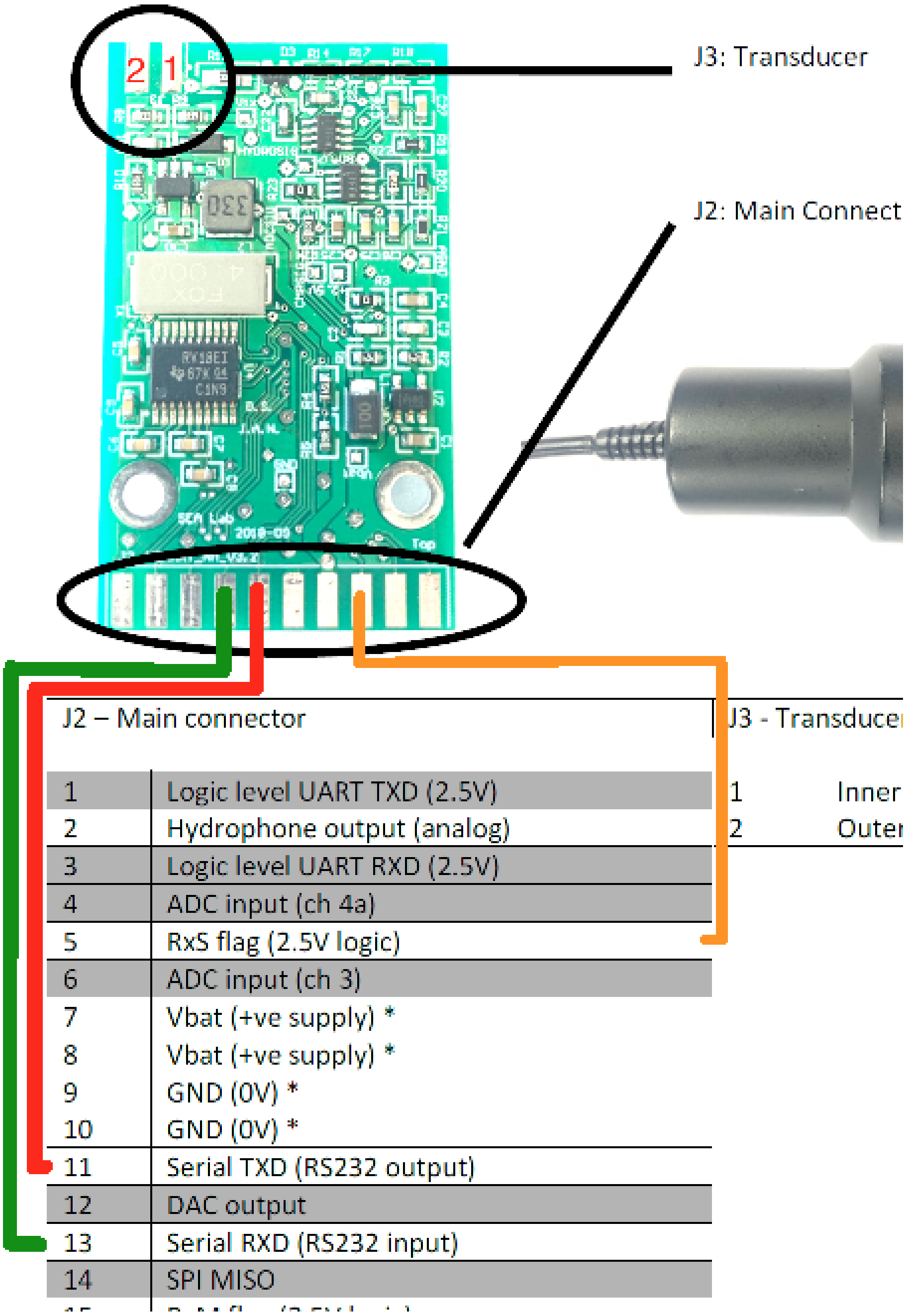
Note: Ensure to heat your potting solution to 45 degrees for an hour to give a nice smooth (no bubbles) finish.

Other qustions

Q- Is it ok to run the pcb without a transducer plugged in?
A- Succorfish cannot guarantee nothing will happen, but they have done this multiple times with no issue.

PCB Pinout

See pinout on the PCB below.



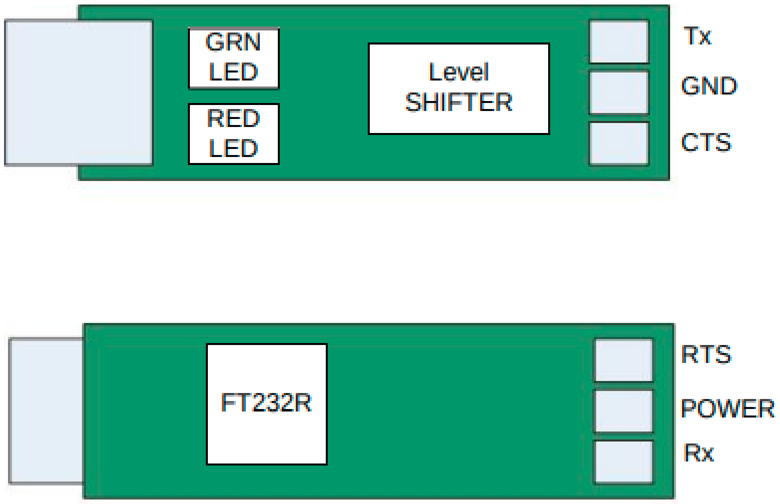
J2 – Main connector	
1	Logic level UART TXD (2.5V)
2	Hydrophone output (analog)
3	Logic level UART RXD (2.5V)
4	ADC input (ch 4a)
5	RxS flag (2.5V logic)
6	ADC input (ch 3)
7	Vbat (+ve supply) *
8	Vbat (+ve supply) *
9	GND (0V) *
10	GND (0V) *
11	Serial TXD (RS232 output)
12	DAC output
13	Serial RXD (RS232 input)
14	SPI MISO

J3 - Transduce	
1	Inner
2	Oute

USB Adapter

The RS232 converter has the following pinout.

To get power on V_out, R3 needs to be bridged.



USB-RS232-PCB Rev 2

Serial Commands

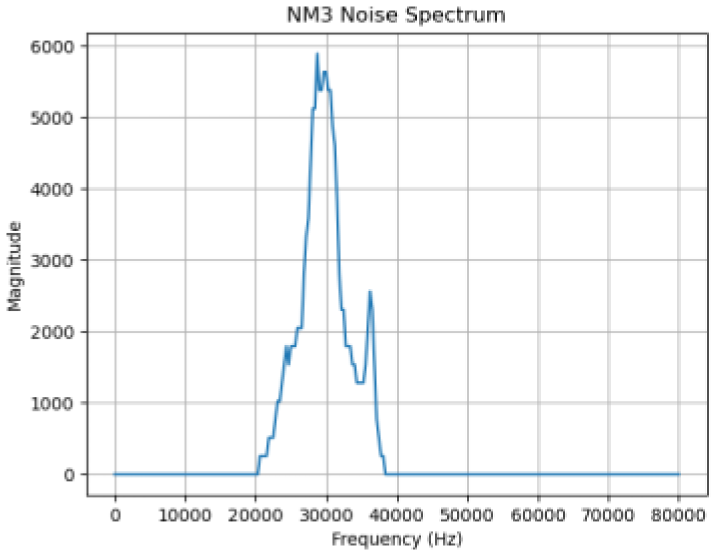
These commands (not complete list) were extracted from the NM3 driver repo and include things like noise magnitude, Fourier spectrum and multipath measurements **not** available in the user guide.

Command	**Serial Code**	**Purpose**	**Typical Usage / Parameters**
—————————	————————————	—————————————————————————	————————————————————————-
Query Status	**$?**	Reads the modem’s status including address, battery voltage, version.	`“$?”` → response with address, voltage, version strings.
Set Address	**$A**	Change the modem’s network address.	`“$Axxx”` – set to numeric address.
Broadcast Data	**$B**	Send a broadcast message.	`“$Bnnddd”` – no ACK expected.
Channel Impulse Response	**$C**	Request impulse response from a remote node (Binary). This can be used to estimate multipath effects and phases.	`“$CMxxx”` (magnitude) or `“$CCxxx”` (complex).
Noise Measurement	**$N**	Measure ambient noise.	`“$N”` → returns **R**MS, **P**eak-to-Peak, **m**ean magnitude e.g. #NR000132P001089M000106
Ping	**$P**	Acoustic ping to remote.	`“$Pxxx”` → returns time of arrival.
Spectrum Measurement	**$S**	Measure local noise spectrum.	`“$S”` → returns number of FFT bins(ASCII) and values(Binary, won't be read with **screen**). Bin frequencies of bin k is likely freq[k] = k × 160000 / 512
Unicast with Ack	**$M**	Send a unicast message and expect ACK.	`“$Mxxxnndddd”` (+ optional timeout/time tag).
Unicast (no ACK)	**$U**	Send a unicast message only.	`“$Uxxxnnddd”`.
System Timer Enable	**$XT** *(Enable/Disable variants)*	Manage system timestamp counter for timed tx/rx.	`“$XTEN”` to enable, `“$XTDI”` to disable, `“$XTCL”` to clear, etc.
~~Get Timer Status~~	~~**$XT?**~~	~~Read current timer status and count.~~	~~`“$XT?”`~~.
Delayed Broadcast	**$B** with timestamp	Same as `$B` but includes a *time tag* for future transmit.	`“$Bddd<timestamp_count>(NOT ASCII)“`.
Delayed Unicast	**$U** with timestamp	As above for unicast.	`”$Uxxxddd,<timestamp_count>(NOT ASCII)“`.

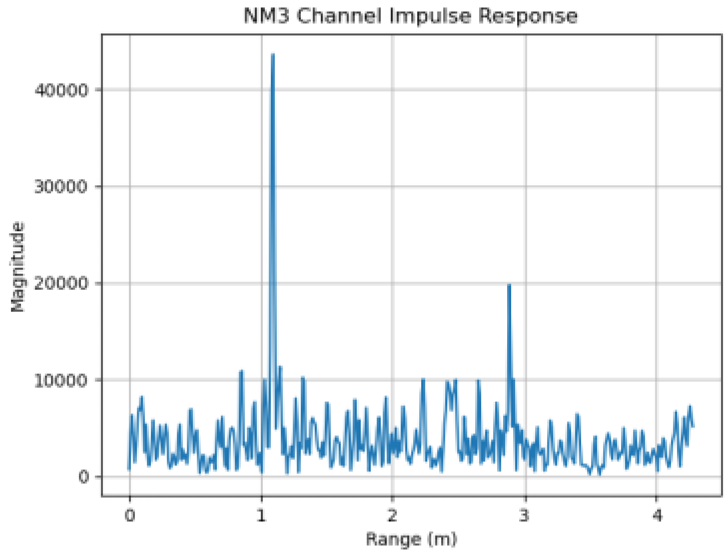
Delayed Unicast Ack	**$M** with timestamp	As above for unicast with ACK.	`”$Mxxxddd<timeout>,<timestamp_count>(NOT ASCII)“`.

Diagnostics

Here's what a properly parsed output of the **$S** command can look like, a Noise Spectral Density where we can se the gain window of the modem between 20-40 kHz.:



And here's the output of the **$CM** command, a simplified Channel Impulse Response where the 1m peak correponds to the direct line and a reflected path arrives with a 2m longer path (5ms in air):



Using a scope we can check the time response of the modem instructed by a microcontroller. Here are some delay values sending a command to the NM3 over serial (9600) from the Teensy 4.1:

From → To	Delay
teensy.write → Transmission start	17 ms
Ping End, modem 1 → Ping answer, modem 2	8 ms
Ping Start, modem 1 → Flag raised, modem 2	30 ms